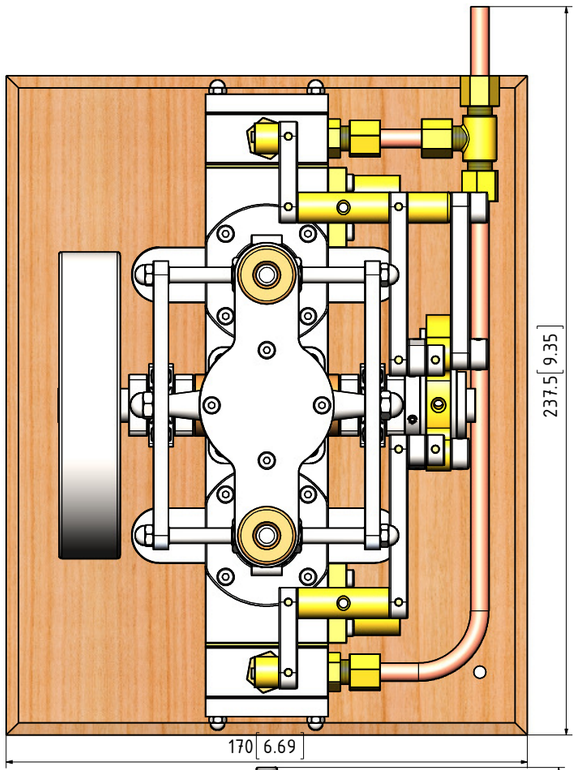
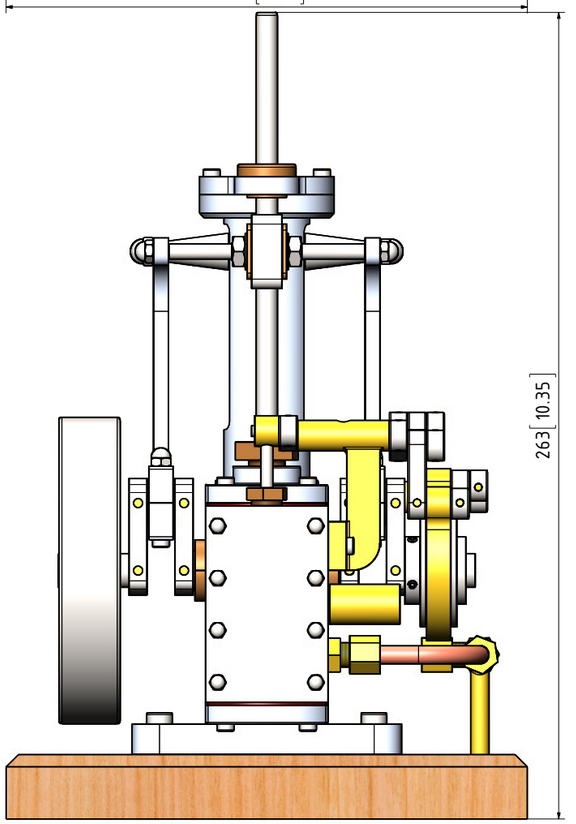
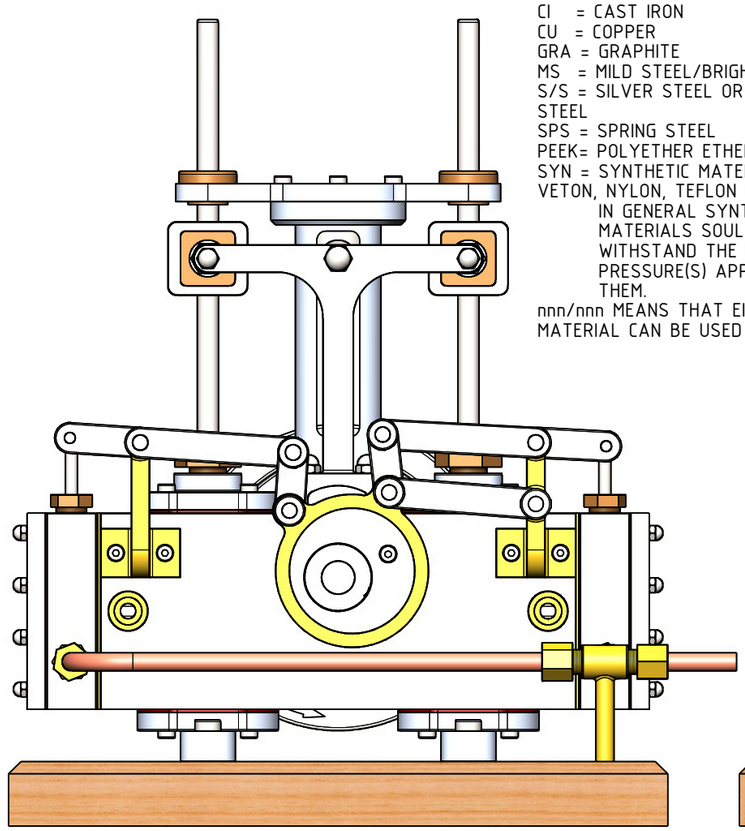
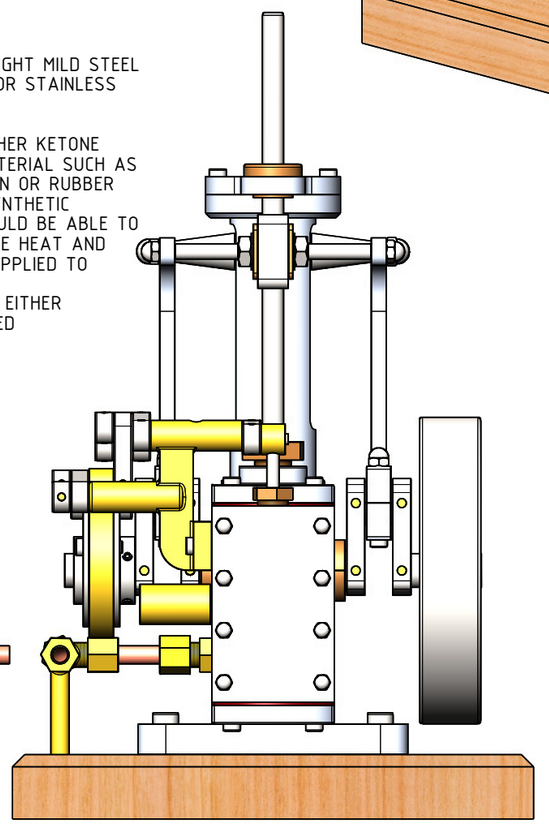






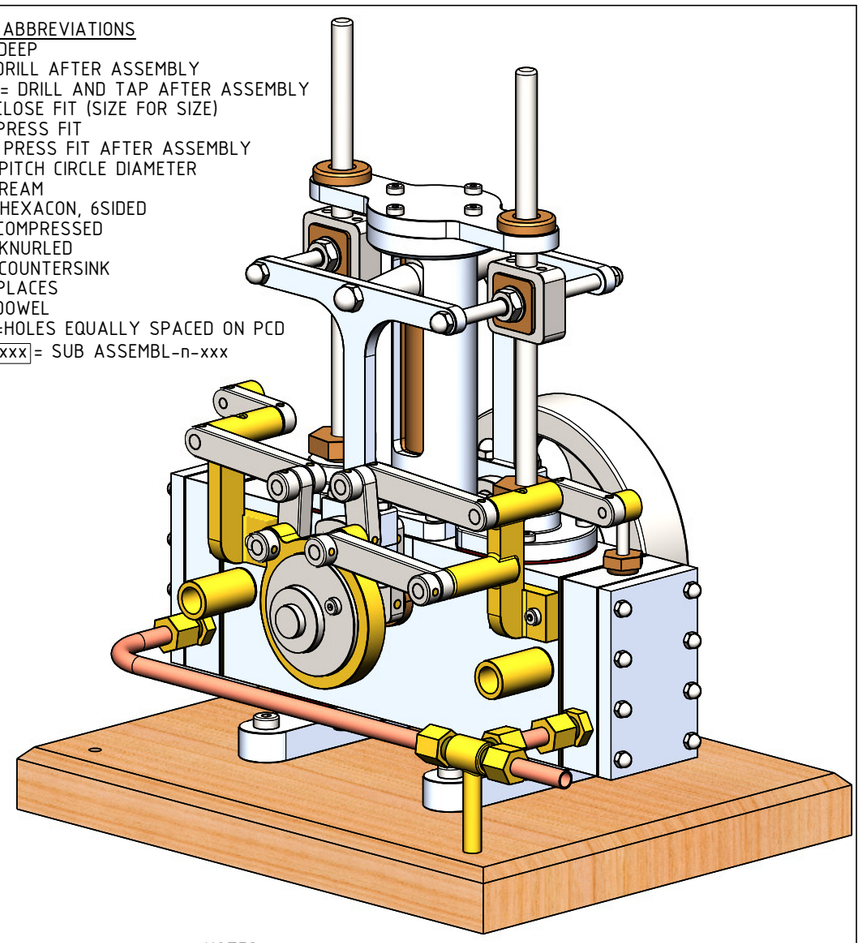


QTY.	PART NUMBER
1	BERNAY-01-WOOD BASE
2	BERNAY-02-CYLINDER BLOCK SUPPORT FOOT
1	BERNAY-03-CYLINDER BLOCK
2	BERNAY-04-CRANK SHAFT BEARING
1	BERNAY-05-CYL. BEARING TOP BLOCK
2	BERNAY-06-EXHAUST PIPE CONNECTOR
4	BERNAY-07-CYLINDER BLOCK GASKET
2	BERNAY-08-CYLINDER LOWER COVER
2	BERNAY-09-CYLINDER UPPER COVER
2	BERNAY-10-PISTON ROD GLAND NUT
1	BERNAY-11-CROSSHEAD GUIDE
1	BERNAY-12-PISTON ROD GUIDE PLATE
2	BERNAY-13-PISTON ROD GUIDE BUSH
2	BERNAY-14-STEAM CHEST
2	BERNAY-15-SLIDE VALVE PORT PLATE
4	BERNAY-16-STEAM CHEST GASKET
2	BERNAY-17-STEAM CHEST COVER
2	BERNAY-18-INLET PIPE NIPPLE
2	BERNAY-19-VALVE ROD GLAND NUT
1	BERNAY-20-RH-ROCKER ARM SUPPORT BRACKET
1	BERNAY-21-LH-ROCKER ARM SUPPORT BRACKET
2	BERNAY-22-STEAM INLET PIPE-1
1	BERNAY-23-STEAM INLET PIPE-2
1	BERNAY-24-STEAM INLET T-CONNECTOR
5	BERNAY-25-M8x6 PIPE NUT
2	BERNAY-26-CRANK SHAFT END
1	BERNAY-27-CRANK SHAFT CENTER
2	BERNAY-28-CRANK SHAFT PIN
4	BERNAY-29-CRANK SHAFT WEB
1	BERNAY-30-FLYWHEEL
1	BERNAY-31-ECCENTRIC SHEAVE
1	BERNAY-32-ECCENTRIC SHEAVE COVER
2	BERNAY-33-PISTON
2	BERNAY-34-PISTON ROD
1	BERNAY-35-CROSSHEAD

QTY.	PART NUMBER
1	BERNAY-36-CROSSHEAD PIN
2	BERNAY-37-CONNECTING ROD SHAFT
2	BERNAY-38-CONNECTING ROD
4	BERNAY-39-PISTON ROD SLIDER
1	BERNAY-40-ECCENTRIC STRAP
2	BERNAY-41-SLIDE VALVE
2	BERNAY-42-SLIDE VALVE ROD
4	BERNAY-43-VALVE ROD RETAINER BUSH
2	BERNAY-44-VALVE GEAR LINK #1
2	BERNAY-45-VALVE GEAR LINK #2
1	BERNAY-46-VALVE GEAR LINK #3
1	BERNAY-47-VALVE GEAR LINK #4
1	BERNAY-48-VALVE GEAR LINK #5
5	BERNAY-49-RETAINER RING
4	BERNAY-M2.5x3 A-K GRUB SCREW
16	BERNAY-M3 DOME NUT
4	BERNAY-M3x6 A-K GRUB SCREW
16	BERNAY-M3x44 THREADED ROD
8	BERNAY-M4 DOME NUT
5	BERNAY-M4x13 A-K CYL HEAD SCREW
20	BERNAY-M4x15 A-K CYL HEAD SCREW
5	BERNAY-M4x20 A-K CYL HEAD SCREW
4	BERNAY-M4x28 A-K CYL HEAD SCREW
2	BERNAY-M5 DOME NUT
2	BERNAY-M5 NUT
8	BERNAY-M6 NUT
4	BERNAY-M6 WASHER
4	BERNAY-M6x10 A-K CYL HEAD SCREW
4	BERNAY-M6x24 A-K CYL HEAD SCREW

MATERIAL ABBREVIATIONS:
ALU = ALUMINIUM
BRS = BRASS
BRZ = BRONZE OR GUNMETAL (BRZ/GM)
CI = CAST IRON
CU = COPPER
GRA = GRAPHITE
MS = MILD STEEL/BRIGHT MILD STEEL
S/S = SILVER STEEL OR STAINLESS STEEL
SPS = SPRING STEEL
PEEK= POLYETHER ETHER KETONE
SYN = SYNTHETIC MATERIAL SUCH AS VETON, NYLON, TEFLON OR RUBBER IN GENERAL SYNTHETIC MATERIALS SOULD BE ABLE TO WITHSTAND THE HEAT AND PRESSURE(S) APPLIED TO THEM.
nnn/nnn MEANS THAT EITHER MATERIAL CAN BE USED

OTHER ABBREVIATIONS
DP = DEEP
DAA= DRILL AFTER ASSEMBLY
D&TAA= DRILL AND TAP AFTER ASSEMBLY
CF = CLOSE FIT (SIZE FOR SIZE)
PF = PRESS FIT
PFAA= PRESS FIT AFTER ASSEMBLY
PCD = PITCH CIRCLE DIAMETER
RM = REAM
HEX = HEXACON, 6SIDED
CP = COMPRESSED
KNL = KNURLED
CSK = COUNTERSINK
PL = PLACES
DWL= DOWEL
HESOP=HOLES EQUALLY SPACED ON PCD
SA-n-xxx= SUB ASSEMBL-n-xxx



NOTES:
0. ALL DRAWINGS ARE IN METRIC MEASUREMENTS
1. ALL ENGINEERING PRACTICES SHALL BE APPLIED WITH REGARDS TO HOLE AND SHAFT TOLERANCES.
2. WHERE SCREWS OR BOLTS ARE USED THE CLEARANCE HOLES SHALL BE APPROXIMATELY 5% TO 8% LARGER THAN THE MATCHING TAPPED HOLE.
3. PREFERABLY ALL TAPPED HOLES AND MATCHING SCREWS AND/OR BOLTS TO BE METRIC FINE (MF)
4. MATERIALS SPECIFIED ON THE DRAWINGS ARE INDICATIVE ONLY. THE BUILDER CAN MAKE HIS/HER OWN MATERIAL CHOICE.
5. ALL CONNECTIONS/JOINTS WHICH HAVE STEAM PRESSURE APPLIED TO IT SHALL BE SILVER/HARD SOLDERED.
6. COMPRESSION SPRINGS ARE DRAWN IN COMPRESSED STATE (CP), UNCOMPRESSED STATE IS APPROX 40% TO 60% LONGER THEN COMPRESSED STATE.
7. WHERE PREFERRED SCREW OR RIVETED CONNECTIONS CAN BE OMITTED AND PARTS CAN BE BONDED TOGETHER BY USING EITHER HIGH STRENGTH GLUE, EPOXY RESIN, OR SOLDER.
8. PARTS WHICH ARE DIRECTLY EXPOSED TO STEAM AND/OR WATER SHOULD BE CONSTRUCTED USING NON-FERROUS OR NON CORROSIVE MATERIAL SUCH AS BRASS, BRONZE, GUNMETAL, STAINLESS STEEL, COPPER OR MONEL.
9. THE ORDER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED AND THE MODEL IS ASSEMBLED IS ENTIRELY LEFT TO THE BUILDER/MODEL MAKER.
10. A COLOUR SCHEME FOR THIS PROJECT IS ENTIRELY LEFT UP TO THE MODEL MAKER.
11. THE MANNER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED IS ENTIRELY LEFT UP TO THE BUILDER.
12. USE LOCTITE, ON SCREW OR PRESS FIT CONNECTIONS OR SURFACES, WERE DEEMED NECESSARY TO PREVENT PARTS FROM LOOSENING.
XX. ERRORS AND/OR OMISSIONS MAY OCCUR IN THE DRAWINGS, DO NOT HESITATE TO CONTACT ME SO THAT THE ERRORS/OMISSIONS CAN BE RECTIFIED.

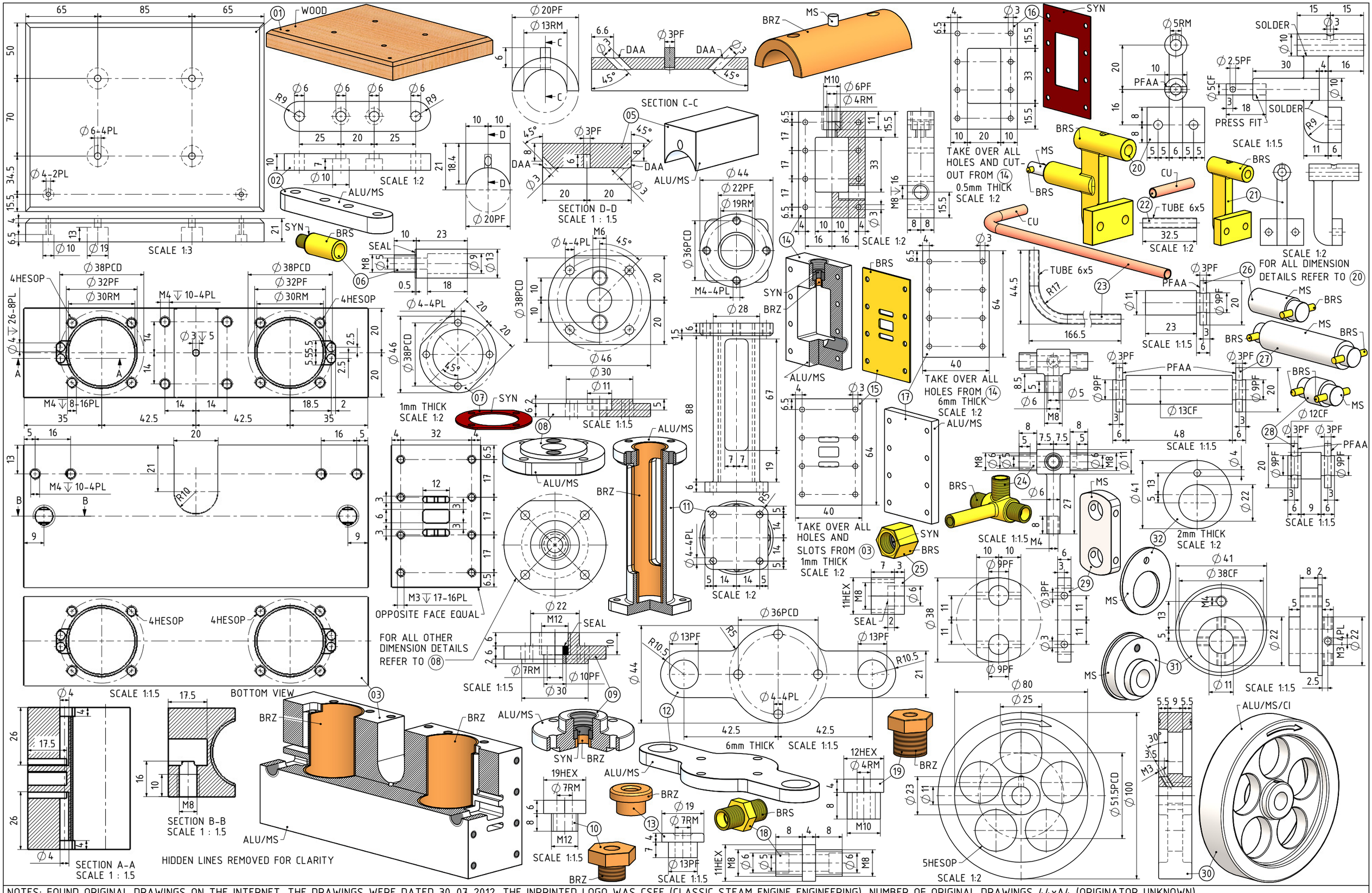
NOTES: FOUND ORIGINAL DRAWINGS ON THE INTERNET. THE DRAWINGS WERE DATED 30-07-2012. THE INPRINTED LOGO WAS CSEE (CLASSIC STEAM ENGINE ENGINEERING). NUMBER OF ORIGINAL DRAWINGS 44xA4 (ORIGINATOR UNKNOWN)

TITLE
2 CYLINDER STEAM ENGINE TO THE BERNAY DESIGN.

DRAWING CONTENTS
GENERAL ARRANGEMENT, ISOMETRIC VIEW, BILL OF MATERIALS, NOTES

JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000
E-MAIL: dewaal@xtra.co.nz.

PROJECT No 09E-19-00
PROJECTION JDWDS
DATE JULY-2016
SHEET: 01 OF 04
MODEL SCALE: 1:1
DWG SCALE: NTS @A3 OR AS SHOWN
Copyright © J.A.M. DE WAAL PAPAKURA NZ
A3 No: BERNAY-01



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TITLE

2 CYLINDER STEAM ENGINE TO THE BERNAY DESIGN.

DRAWING CONTENTS

PARTS AND ASSEMBLIES

PROJECT No 09E-19-00

JDW DRAUGHTING SERVICES

J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000 E-MAIL: dewaal@xtra.co.nz.

PROJECTION

JDWDS

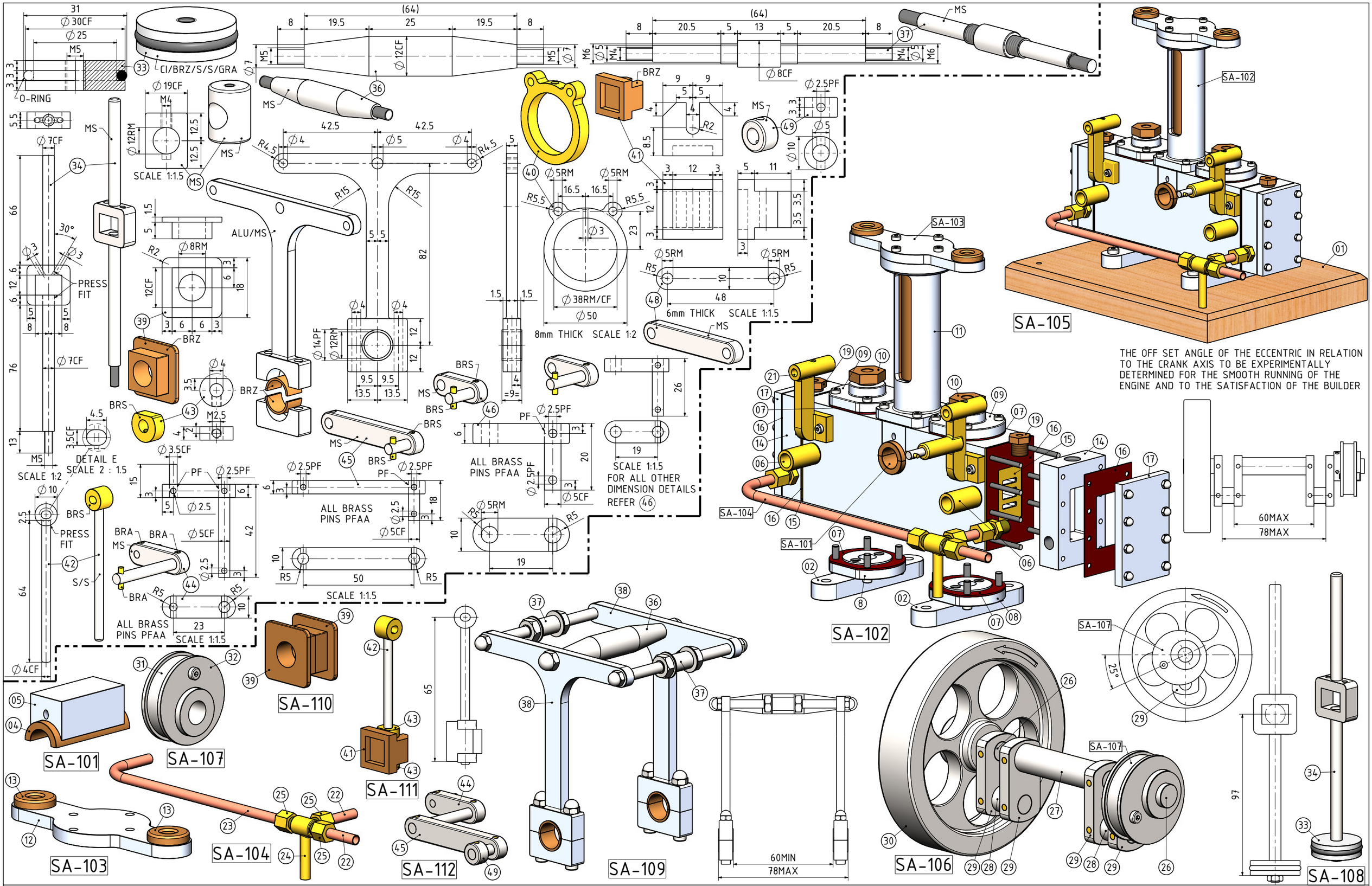
DATE JULY-2016

MODEL SCALE: 1:1

DWG SCALE: 1:1 @A3 OR AS SHOWN

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SHEET: 02 OF 04 | A3 | No: BERNAY-02



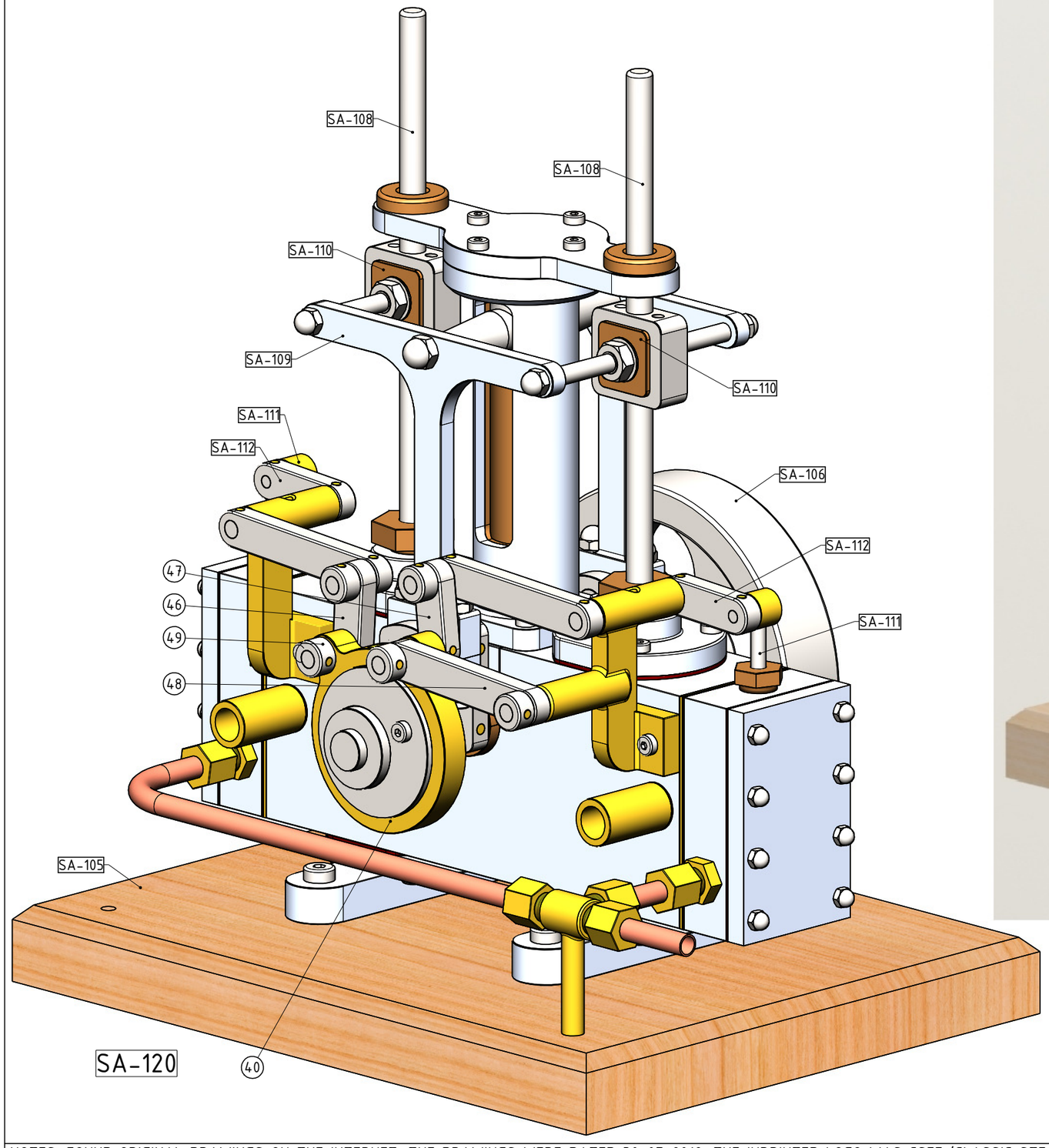
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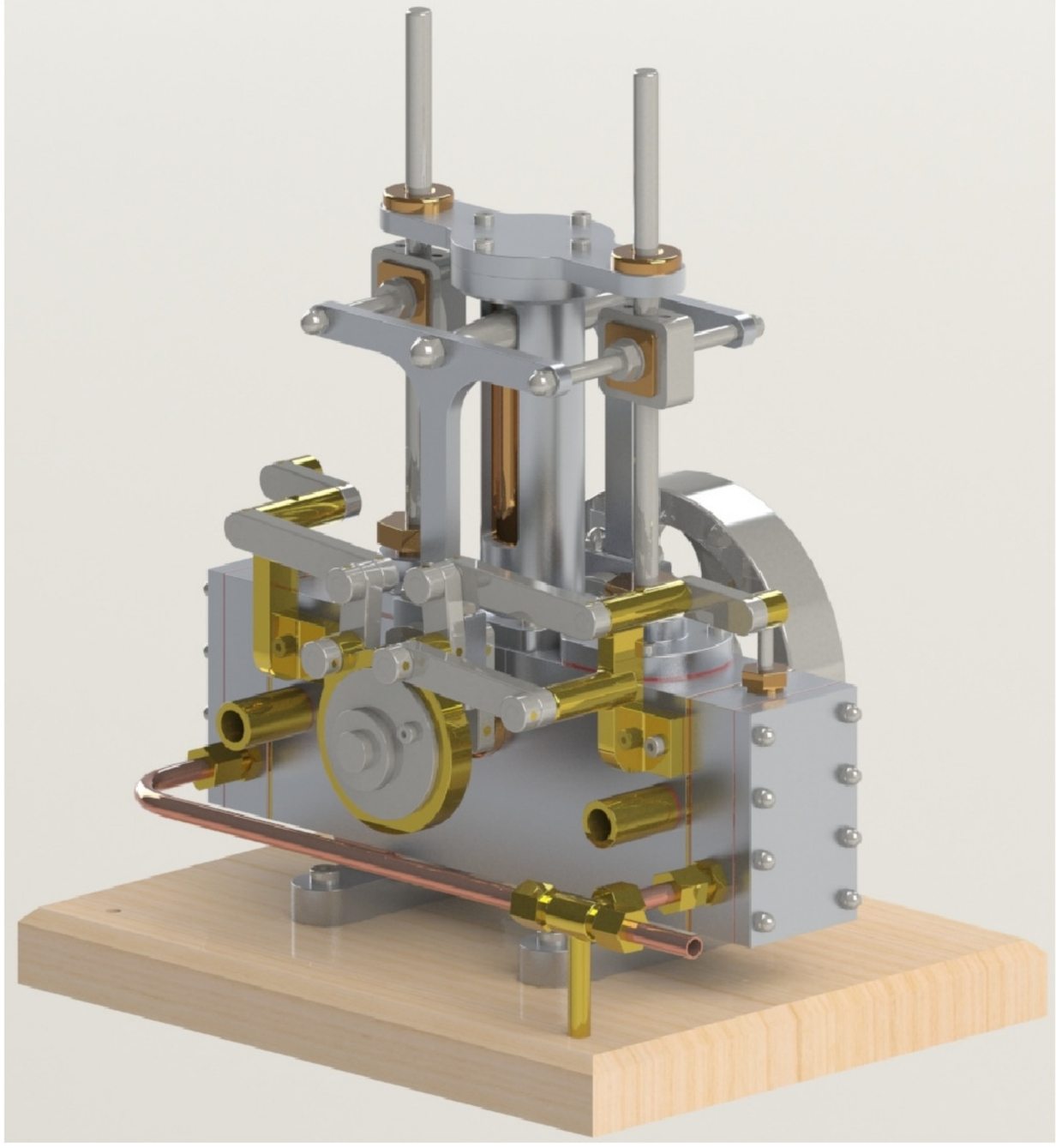
TITLE
2 CYLINDER STEAM ENGINE TO THE BERNAY DESIGN.

DRAWING CONTENTS
PARTS AND ASSEMBLIES

JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000
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PROJECT No 09E-19-00
PROJECTION JDWDS
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SHEET: 03 OF 04
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TITLE
2 CYLINDER STEAM ENGINE TO THE BERNAY DESIGN.

DRAWING CONTENTS
ASSEMBLIES AND RENDERED PICTURE

PROJECT No 09E-19-00
JDW DRAUGHTING SERVICES
J.A.M. DE WAAL. 12 BRIGHTWELL STREET PAPAKURA 2110. NEW ZEALAND. PHONE: 0064 09 2988815. MOB: 0211791000
E-MAIL: dewaal@xtra.co.nz.

PROJECTION JDWDS
DATE JULY-2016
SHEET: 04 OF 04

MODEL SCALE: 1:1
DWG SCALE: 1:1 @A3 OR AS SHOWN
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A3 No: BERNAY-04